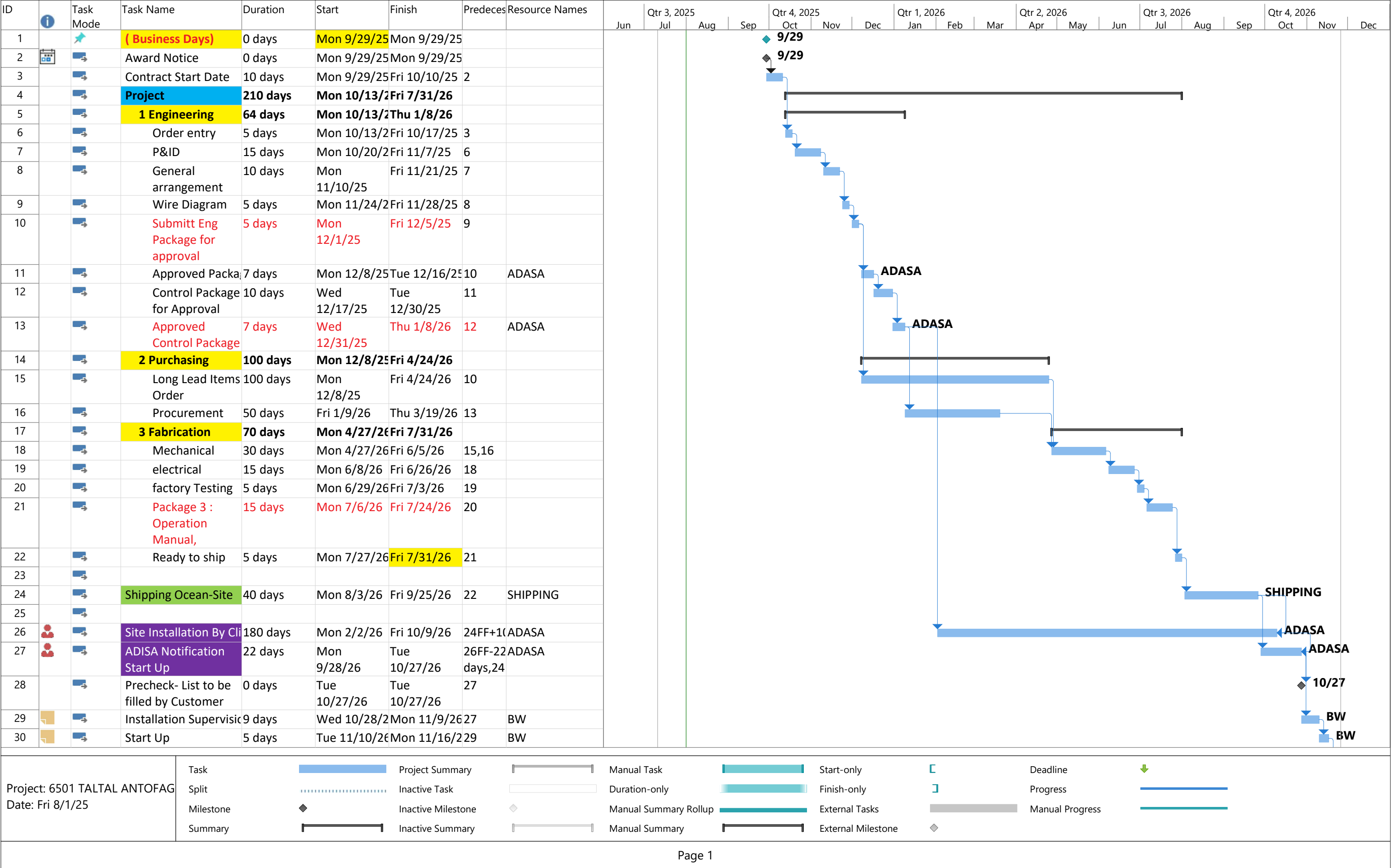


Technical Proposal

Engineering Submittals						
References: G = General P = Process M = Mechanical E = Electrical I&C = Instrumentation & Control Q = Quality (Assurance / Control)						
Doc. No.	Discipline	Document Name	Submittal Due			Notes
			#	Unit	Event	
TBA-16		Detailed Bill of Materials (BOL), with brands and models	within 90	Days	after the start date	
TBA-17		3D Model	within 90	Days	after the start date	
TBA-18		Isometric views of high-pressure lines.	within 90	Days	after the start date	
TBA-19		Proposal of rail beams and internal lifting points within the module for the removal of mechanical equipment (pumps and turbochargers).	within 90	Days	after the start date	
TBA-20	Q	Detailed Inspection and Testing Plan (ITP) (in Spanish PIE), as per requirements defined in document P22-IT-09-000-001-0 (this is minimum basis)	within 90	Days	after the start date	
TBA-21	G	Spare Parts and Special Tool List	1	Month	Before shipment	
TBA-22		Others: -) A/C report (a thermal calculation report to determine the volume of the air conditioning system)	within 90	Days	after the start date	
TBA-23	G	Assembly/Installation Instruction Manual	1	Month	Before shipment	
TBA-24	G	Start-up Manual of the System-Plant /Equipment	1	Month	Before shipment	
TBA-25	G	Operation and Maintenance (O&M) Manual of the System-Plant /Equipment	1	Month	Before shipment	

Technical Proposal

Engineering Submittals						
<u>References:</u> G = General P = Process M = Mechanical E = Electrical I&C = Instrumentation & Control Q = Quality (Assurance / Control)						
Doc. No.	Discipline	Document Name	Submittal Due			Notes
			#	Unit	Event	
TBA-26		Final Software and License Package	1	Month	Before shipment	
TBA-27	G	Detailed Procedure for Preservation, Packaging, and Preparation for International Maritime Transport	1	Month	Before shipment	

Start-onlyFinish-onlyExternal TasksExternal MilestoneDeadlineProgressManual Progress

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ID	i	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names	Qtr 3, 2025				Qtr 4, 2025			Qtr 1, 2026			Qtr 2, 2026			Qtr 3, 2026			Qtr 4, 2026		
									Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
31			Training	2 days	Tue 11/17/25	Wed 11/18/25	30	BW																			
32			Performance Test	5 days	Thu 11/19/25	Wed 11/25/25	31	BW																			

Project: 6501 TALTAL ANTOFAG

Date: Fri 8/1/25

Task

Split

Milestone

Summary

Project Summary

Inactive Task

Inactive Milestone

Inactive Summary

Manual Task

Duration-only

Manual Summary Rollup

Manual Summary

Start-only

Finish-only

External Tasks

External Milestone

Deadline

Progress

Manual Progress

Page 2

Technical Proposal

12. INSPECTION AND TESTING PLAN (ITP)

(Ref.: P22-ET-09-000-001-0, Section 6 -Item “ PIE (Plan de Inspección y Ensayos Detallado (PIE) de equipos principales según Section 5.1”)

Inspection and Testing Plan (ITP) sample is attached for reference only. The final document for this project will be developed and submitted after the contract is awarded.

		INSPECTION & TESTING PLAN - SAMPLE						
		CUSTOMER:	Aguas Antofagasta					REV.: 0
		LOCATION:	Chile					DATE:
		PROJECT No.:	Proposal 20.24.6501 -ADASA Brine Recovery Project -UHPRO System					ORIG. BY:
No.	Item Name	Description of Inspection & Test		Reference Document	Verification			Inspection Doc.
					MFG	BWWA	Owner	
1	Spools - Stainless Steel & Super Duplex	10	Verification of Material	Material Spec according to ASTM Code	R	R	R	Mill Certs. or Test Report
			- Material Test Report Review/Certificate					
		20	Non Destructive Examination (NDE)	ASME VIII	I	R	R	Inspection Report
		30	Visual & Dimensional Check	Drawings / Prints	I	R	R	Inspection Report
		40	Pickle / Passivation or Electropolish	ASTM A967-05	I	R	R	Manufacturer Procedure
		50	Hydrostatic test	ASME B31.3	I	R	R	Test Report
		60	Packing Inspection	Manufacturer Standard Procedure	I	R	R	Packing List &Photos
			Marking Condition / Protection for shipment					
		70	Documentation :	Drawings/ Print , Technical spec , and Specifications	I	-	-	Data Book
			- Quality Record Book					
			- Certificate of Compliance (if required)					
			- Data Book Review / Quality Release					
2	Non-metal Valves PVC, CPVC	10	Compliance to requirements	Manufacturer Standard Procedure	R	R	R	Purchase Order
		20	Packaging Inspection	Manufacturer Standard Procedure	I	R	R	Photos
			Marking , Quantity, Packing					
		30	Documentation :	Drawings/ Print , Technical spec , and Specifications	I	-	-	Data Book
			- Quality Record Book					
			- Certificate of Compliance (if required)					
			- Data Book Review / Quality Release					
3	Metal Valves : Solenoid, Manual & Pneumatic	10	Verification of Material	Material Spec according to ASTM Code	R	R	R	Mill Certs. or Test Report
			- Material Test Report Review/Certificate					
		20	Visual & Dimension Inspection	Approved Drawing(s)	I	R	R	Supplier's dimensional inspection report
		30	Testing:	Manufacturer Standard Procedure	I	R	R	Factory certificate
			- Seat Leakage Test					
			- Body Pressure Test					

 BW WATER		INSPECTION & TESTING PLAN - SAMPLE								
		CUSTOMER:		Aguas Antofagasta					REV.:	0
		LOCATION:		Chile					DATE:	
		PROJECT No.:		Proposal 20.24.6501 -ADASA Brine Recovery Project -UHPRO System						ORIG. BY:
No.	Item Name	Description of Inspection & Test		Reference Document	Verification			Inspection Doc.		
					MFG	BWWA	Owner			
		40	Operation Test	Manufacturer Standard Procedure	I	R	R	Test Certificate		
		50	Painting Inspection (if applicable):	Manufacturer Standard Procedure	I	R	R	Inspection Report		
			- Color / Visual Condition Check							
		60	Packing Inspection	Manufacturer Standard Procedure	I	R	R	Packing List &Photos		
			Marking Condition / Protection for shipment							
		70	Documentation :	Drawings/ Print , Technical spec , and Specifications	I	-	-	Data Book		
			- Quality Record Book							
			- Certificate of Compliance (if required)							
			- Data Book Review / Quality Release							
		4	Centrifugal Pumps and Motors	10	Material Verification :	Material Spec according to ASTM code & Manufacturer Standard Procedure	R	R	R	Mill Certs. or Report
Material Test report Review										
20	Motor Test & Inspection :			Data Sheet	I	R	R	Test Report		
30	Hydrostatic Test			Manufacturer Standard Procedure	I	R	R	Test Procedure		
40	Balance Test (For Impeller or Rotor)			Manufacturer Standard Procedure	I	R	R	Test Report		
50	Dimensional and visual inspection			Approved drwg, data sheet	I	R	R	Dim. Inspection		
60	Performance & Running Test :			Manufacturer Standard Procedure	I	R	R	Test Report		
70	Painting Inspection :			Manufacturer Standard Procedure	I	R	R	Painting Report		
80	Documentation :			Drawings/ Print , Technical spec and Specifications	I	-	-	Data Book		
	- Quality Record Book									
	- Certificate of Compliance (if required)									
	- Data Book Review / Quality Release									
5	Metering, Unloading & Dispensing Pumps(COTS):			10	Compliance to requirements	Per Manufacturer Standard Procedure & Datasheet	R	R	R	Purchase Order
		- Pump test certificate								
		20	Visual & Dimension Inspection	Data Sheet & Manufacturer Standard Procedure	I	R	R	Data Sheet		
		30	Documentation :	Drawings/ Print , Technical spec , and Specifications	I	-	-	Data Book		
			- Quality Record Book							

		INSPECTION & TESTING PLAN - SAMPLE						
		CUSTOMER:	Aguas Antofagasta					REV.: 0
		LOCATION:	Chile					DATE:
		PROJECT No.:	Proposal 20.24.6501 -ADASA Brine Recovery Project -UHPRO System					ORIG. BY:
No.	Item Name	Description of Inspection & Test		Reference Document	Verification			Inspection Doc.
					MFG	BWWA	Owner	
		50	- Certificate of Compliance	Drawings/ Print , Technical spec , and Specifications				Data Book
			- Data Book Review / Quality Release					
6	RO Elements	10	Testing	Data sheets	R	R	R	Test Certificates
		20	Dimensional and visual inspection	Approved drwg, data sheet	I	R	R	Certificate of Conformance
		30	Packing Inspection	Manufacturer Standard Procedure	I	R	R	Packing List &Photos
			Marking Condition / Protection for shipment					
		40	Documentation :	Drawings/ Print , Technical spec , and Specifications	I	-	-	Data Book
			- Quality Record Book					
			- Certificate of Compliance (if required)					
			- Data Book Review / Quality Release					
7	RO Pressure Vessels (FRP)	10	Material Verification	ASTM D 3299 / D4097	I	R	R	Material certificates, test reports
			- Material Test Report Review					
		20	Dimensional and visual inspection	Approved drwg and DHT MPS	I	R	R	Supplier's process
		30	Nameplate/markings verification	Approved DWG. and MPS	I	R	R	Supplier's dimensional inspection report
		40	Hydrostatic Test	Test certification to ASME X. Pressure rating:	I	R	R	Test Report
				- 1000 PSI for SWRO,				
				- 300 PSI BWRO.				
		50	Painting Inspection	Manufacturer Standard Procedure	I	R	R	Painting Report
		60	Packaging Inspection	Manufacturer Standard Procedure	I	R	R	Packing list and photos
			- Marking , Quantity, Packing					
		70	Documentation :	Drawings/ Print , Technical spec , and Specifications	I	-	-	Data Book
			- Quality Record Book					
			- Certificate of Compliance (if required)					
			- Data Book Review / Quality Release					
10	Energy Recovery Device	10	Verification of Material	Test Certificate	R	R	R	Test Certificate

		INSPECTION & TESTING PLAN - SAMPLE						
		CUSTOMER:	Aguas Antofagasta					REV.: 0
		LOCATION:	Chile					DATE:
		PROJECT No.:	Proposal 20.24.6501 -ADASA Brine Recovery Project -UHPRO System					ORIG. BY:
No.	Item Name	Description of Inspection & Test		Reference Document	Verification			Inspection Doc.
					MFG	BWWA	Owner	
		20	Dimensional and visual inspection	Approved drwg, data sheet	I	R	R	Dim. Inspection
		30	Factory or Performance Test	Data Sheet & Manufacturer Standard Procedure	I	R	R	Test Report
		40	Packing Inspection	Manufacturer Standard Procedure	I	R	R	Photos
			- Marking Condition / Protection for shipment					
		50	Documentation :	Drawings/ Print , Technical spec , and Specifications	I	-	-	Data Book
			- Quality Record Book					
			- Certificate of Compliance (if required)					
			- Data Book Review / Quality Release					
8	Tanks -CIP	10	Material Verification	Material Spec according to ASTM code	R	R	R	Mill Certs. or Report according to EN 10204
		20	Visual and Dimensional Inspection for Assembly	Drawing	I	R	R	Dim. Inspection Report
		30	Non-Destructive Examination (NDE) (if applicable)	ASME Section VIII	I	R	R	PT Report
		40	Hydrostatic Test -	Manufacturer Standard Procedure	I	R	R	Hydro Test Report
			- Full Water Filling Test					
		50	Tank Lining Verification: (If applicable)	Manufacturer Standard Procedure	I	R	R	Manufacturer Procedure
			- Cleaning Hardness		I	R	R	
			- Thickness		I	R	R	
			- Pinhole Test		I	R	R	
		60	Painting Inspection (if applicable)	Manufacturer Standard Procedure	I	R	R	Painting Report
			- Dry Film Thickness		I	R	R	
			- Color / Visual Condition Check		I	R	R	
		70	Packaging Inspection	Manufacturer Standard Procedure	I	R	R	Packing List &Photos
			- Marking , Quantity, Packing		I	R	R	
		80	Documentation :	Drawings/ Print , Technical spec and Specifications	I	-	-	Data Book
			- Quality Record Book					
			- Certificate of Compliance (if required)					
			- Data Book Review / Quality Release					

		INSPECTION & TESTING PLAN - SAMPLE						
		CUSTOMER:	Aguas Antofagasta					REV.: 0
		LOCATION:	Chile					DATE:
		PROJECT No.:	Proposal 20.24.6501 -ADASA Brine Recovery Project -UHPRO System					ORIG. BY:
No.	Item Name	Description of Inspection & Test		Reference Document	Verification			Inspection Doc.
					MFG	BWWA	Owner	
9	Skid / Steel Structure (CS)	10	Visual & Dimensional Check	Approved drwg, data sheet	I	R	R	Dim. Inspection
		20	Hydro Test for all pipes	Manufacturer Standard Procedure	I	R	R	Hydro Test Certificate
		30	Assembly Inspection	Approved drwg, BOM, Mfg Standard Procedure	I	R	R	0
		50	Painting Inspection (if applicable)	Manufacturer Standard Procedure	I	R	R	Painting Report
			- Dry Film Thickness					
			- Color / Visual Condition Check					
		60	Packaging Inspection	Manufacturer Standard Procedure	I	R	R	Packing List &Photos
			- Marking , Quantity, Packing					
		70	Documentation :	Drawings/ Print , Technical spec and Specifications	I	-	-	Data Book
			- Quality Record Book					
			- Certificate of Compliance (if required)					
			- Data Book Review / Quality Release					
10	RO Skids	10	Test Report Review		R	R	R	Data Sheet
		20	Dimensional and visual inspection	Approved drwg, data sheet	I	R	R	Dim. Inspection
		30	Hydrostatic Test	Manufacturer Standard Procedure	I	R	R	Test Report
		40	Painting Inspection	Manufacturer Standard Procedure	I	R	R	Painting Report
			- Color / Visual Condition Check on final coat only.					
		50	Packaging Inspection	Manufacturer Standard Procedure	I	R	R	Packing List &Photos
			- Marking , Quantity, Packing					
		60	Documentation :	Drawings/ Print , Technical spec , and Specifications	I	-	-	Data Book
			- Quality Record Book					
			- Certificate of Compliance (if required)					
			- Data Book Review / Quality Release					
11	Chemical Dosing Systems	10	Dimension Inspection	Drawing	I	R	R	Supplier's dimensional inspection report
		20	Hydrostatic Test	Manufacturer Standard Procedure	I	R	R	Test Report

		INSPECTION & TESTING PLAN - SAMPLE						
		CUSTOMER:	Aguas Antofagasta					REV.: 0
		LOCATION:	Chile					DATE:
		PROJECT No.:	Proposal 20.24.6501 -ADASA Brine Recovery Project -UHPRO System					ORIG. BY:
No.	Item Name	Description of Inspection & Test		Reference Document	Verification			Inspection Doc.
					MFG	BWWA	Owner	
		30	Painting Inspection (If applicable)	Manufacturer Standard Procedure	I	R	R	Painting Report
			- Color / Visual Condition					
			Check on final coat only.					
		40	Packaging Inspection	Manufacturer Standard Procedure	I	R	R	Packing List &Photos
			- Marking , Quantity, Packing					
		50	Documentation :	Drawings/ Print , Technical spec , and Specifications	I	-	-	Data Book
			- Quality Record Book					
			- Certificate of Compliance (if required)					
			- Data Book Review / Quality Release					
12	Local Instrumentation & Electrical Raw Material	10	Dimensional and visual inspection	Approved drwg, data sheet	I	R	R	Dim. Inspection
			- Record actual dimensions on inspection report (for non-COTS or Catalog items)					
		20	Testing (If applicable) ;	Manufacturer Standard Procedure	I	R	R	Cut sheet
			- Calibration or Functional Test (for non-COTS or Catalog items)					
		30	Packaging Inspection	Manufacturer Standard Procedure	I	R	R	Packing List &Photos
			- Marking , Quantity, Packing					
		40	Documentation :	Drawings/ Print , Technical spec , and Specifications	I	-	-	Data Book
			- Quality Record Book					
			- Certificate of Compliance (if required)					
			- Data Book Review / Quality Release					
13	Electric Heaters:	10	Visual and Dimensional Inspection for Assembly	Drawing	I	R	R	Dim. Inspection Report
		20	Function Test / Compliance to requirements	Manufacturer Standard Procedure	R	R	R	Data Sheet
		30	Packaging Inspection		I	R	R	Packing List &Photos
			- Marking , Quantity, Packing					
		40	Documentation :	Drawings/ Print , Technical spec , and Specifications	I	-	-	Data Book
			- Quality Record Book					
			- Certificate of Compliance (if required)					

		INSPECTION & TESTING PLAN - SAMPLE								
		CUSTOMER:		Aguas Antofagasta				REV.:	0	
		LOCATION:		Chile				DATE:		
		PROJECT No.:		Proposal 20.24.6501 -ADASA Brine Recovery Project -UHPRO System				ORIG. BY:		
No.	Item Name	Description of Inspection & Test		Reference Document	Verification			Inspection Doc.		
					MFG	BWWA	Owner			
			- Data Book Review / Quality Release							
14	Control Panel	10	Visual & Dimensional Check	Approved drwg, data sheet	I	R	R	Dim. Inspection		
		20	Wiring and Assembling	Drawing	I	R	R	Drawing		
		30	Insulation Resistance Test	Manufacturer Standard Procedure	I	R	R	Manufacturer Procedure		
		40	Painting Inspection (if applicable)	Manufacturer Standard Procedure	I	R	R	Painting Report		
			- Dry Film Thickness							
			- Color / Visual Condition Check							
		50	Packaging Inspection	Manufacturer Standard Procedure	I	R	R	Packing List &Photos		
			- Marking , Quantity, Packing							
		60	Documentation :	Drawings/ Print , Technical spec , and Specifications	I	-	-	Data Book		
			- Quality Record Book							
			- Certificate of Compliance (if required)							
			- Data Book Review / Quality Release							
		#	Iteam name xxx	10	Xxxx	xxx	XX	XX		xx
				20	xxxx	xxx				xx

Technical Proposal

13. JUSTIFICATION OF MECHANICAL COUPLINGS/JOINTS (VICTAULIC TYPE) IN HIGH PRESSURE

(Ref.: P22-ET-09-000-001-0, Section 6 -Item “Justificación de Tipos de Junta en Alta Presión”)

HIGH PRESSURE LINE JOINT LIST

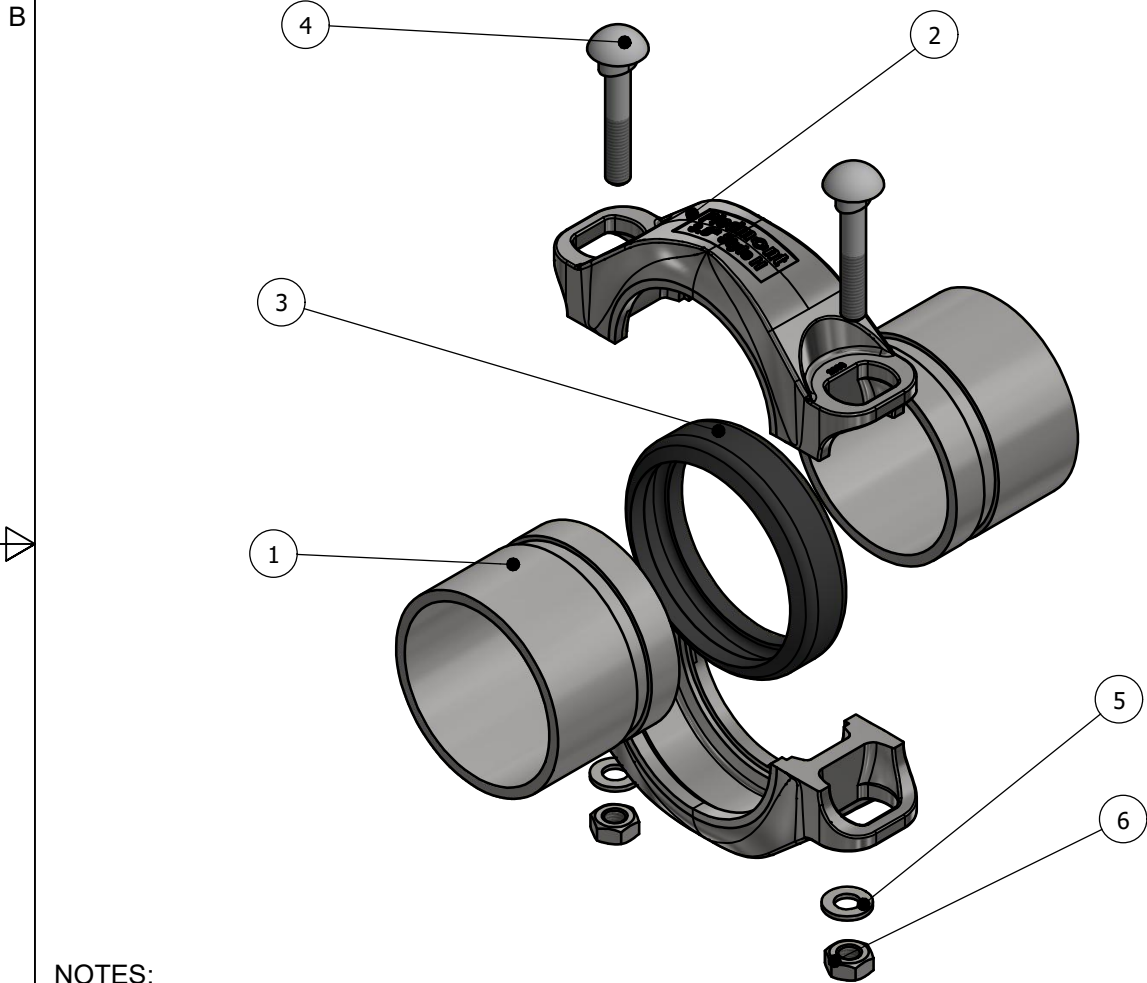
No.	Service	Process condition	Joint Types	Process Pipe size	Pipe Material	Pipe Class	Remark
1	Discharge from RO HP Feed Pump (High Pressure Line)	49 m3/hr 48.8 Barg pH 7.6 19 - 24 °C	Victaulic coupling or Welded or flanged Joints(CL900)	DN 80, NPS 3"	SDSS	CL 900	For stability and durability of HP pipe 1) Victaulic couplings (Piedmont): Rated as 2000PSI (138 BAR). 2) Flanged or Welded connection: Test can be done up to CL 900 (2220PSIG) for pipe spools in HP line
2	Feed into UHPRO Train (Feed Turbo HP OUT), (High Pressure Line)	49 m3/hr 67.5 Barg pH 7.6 19 - 24 °C	Victaulic coupling or Welded or flanged Joints(CL900)	DN 80, NPS 3"	SDSS	CL 900	For stability and durability of HP pipe 1) Victaulic couplings (Piedmont): Rated as 2000PSI (138 BAR). 2) Flanged or Welded connection: Test can be done up to CL 900 (2220PSIG) for pipe spools in HP line
3	1st Stage Concentrate of UHPRO, (High Pressure Line)	35.1 m3/hr 66.3 Barg pH 7.6 19 - 24 °C	Victaulic coupling or Welded or flanged Joints(CL900)	DN 65, NPS 2.5"	SDSS	CL 900	For stability and durability of HP pipe 1) Victaulic couplings (Piedmont): Rated as 2000PSI (138 BAR). 2) Flanged or Welded connection: Test can be done up to CL 900 (2220PSIG) for pipe spools in HP line
4	Feed into 2nd Stage UHPRO Train (Interstage Turbo HP OUT) , (High Pressure Line)	35.1 m3/hr 83.9 Barg pH 7.6 19 - 24 °C	Victaulic coupling or Welded or flanged Joints(CL900)	DN 65, NPS 2.5"	SDSS	CL 900	For stability and durability of HP pipe 1) Victaulic couplings (Piedmont): Rated as 2000PSI (138 BAR). 2) Flanged or Welded connection: Test can be done up to CL 900 (2220PSIG) for pipe spools in HP line

Technical Proposal

HIGH PRESSURE LINE JOINT LIST							
No.	Service	Process condition	Joint Types	Process Pipe size	Pipe Material	Pipe Class	Remark
5	2nd Stage Concentrate of UHPRO or Feed into Interstage Turbo (Interstage Turbo HP IN), (High Pressure Line)	27.9 m3/hr 82.4 Barg pH 7.6 19 - 24 °C	Victaulic coupling or Welded or flanged Joints(CL900)	DN 50, NPS 2"	SDSS	CL 900	For stability and durability of HP pipe 1) Victaulic couplings (Piedmont): Rated as 2000PSI (138 BAR). 2) Flanged or Welded connection: Test can be done up to CL 900 (2220PSIG) for pipe spools in HP line
6	Feed into Feed Turbo (Feed Turbo HP IN) or Discharge from Interstage Turbo (Interstage LP DISCHARGE), (High Pressure Line)	27.9 m3/hr 51.1 Barg pH 7.6 19 - 24 °C	Victaulic coupling or Welded or flanged Joints(CL900)	DN 50, NPS 2"	SDSS	CL 900	For stability and durability of HP pipe 1) Victaulic couplings (Piedmont): Rated as 2000PSI (138 BAR). 2) Flanged or Welded connection: Test can be done up to CL 900 (2220PSIG) for pipe spools in HP line

Please see attached technical details of PIEDMONT Pacific Groove End Flexible Pipe Coupling

ITEM	DESCRIPTION	MATERIAL	QTY
1	GROOVED END PIPE (AWWA C606 STANDARD)	METALLIC/NON METALLIC	2
2	COUPLING HOUSING	CE8MN/CE3MN (DUPLEX/SUPERDUPLEX STAINLESS STEEL)	2
3	GASKET	EPDM RUBBER	1
4	ROUND HEAD, OVAL NECK TRACK BOLT / HEXAGONAL HEAD BOLT	316 SS/ 2205 DUPLEX	2
5	WASHER	316 SS / 2205 DUPLEX	2
6	HEAVY HEX NUT	SILICON BRONZE/316 SS / 2205 DUPLEX	2



NOTES:

- DIMENSIONS SHOWN IN THE DRAWING ARE INDICATIVE. FOR DETAILED INFORMATION ON GD&T, CONTACT PIEDMONT
- COUPLING IS RATED FOR SCH. 40 PIPE (OR THICKER) WITH CUT GROOVES ONLY
- HOUSING MATERIAL IS DUPLEX (CE8MN) OR SUPER DUPLEX (CE8MN/CE3MN) STAINLESS STEEL, CONFORM TO ASTM A995/A995M-13
- BOLTS ARE ROUND HEAD, OVAL NECK 316SS/2205 DUPLEX TRACK BOLTS
- NUTS ARE HEAVY HEX SILICON BRONZE/316SS / 2205 DUPLEX, CONFORM TO DIN-934.
- WASHERS ARE 316SS/ 2205 DUPLEX,CONFORM TO DIN-125.
- GASKETS ARE EPDM RUBBER (FLUSH-FIT/C-SHAPE)

PROPRIETARY AND CONFIDENTIAL NOTE:
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF PIEDMONT PACIFIC. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF PIEDMONT PACIFIC IS PROHIBITED.

REV		DATE	REVISION DESCRIPTION	DRAWN	CHKD	ENG	APPVD
5		18-07-2016	PRODUCT WEIGHTS ARE ADDED. DIMENSIONS REVISED.	AH	CZZ	AH	AH
6		08-09-2017	TABLE UPDATED	JF	CZZ		
7		02-11-2017	TABLE UPDATED	JF	CZZ		
8		09-08-2018	TABLE UPDATED	JF	CZZ		
9		08-05-2020	TABLE UPDATED / ADDED. 2-1/2 SIZE	DT	AS	AS	DJ

Piedmont

UNLESS NOTED OTHERWISE
INTERPRETATION:
ANSI Y14.5
TOLERANCES: ±0.03" [0.75 MM]
ANGLES: ±1°
HOLE SIZES: ±0.00" [0.75 MM]
HOLE CENTERS: ±0.00" [0.75 MM]
SURFACE FINISH ON PART: 125
SURFACE FINISH ON D-RING: 5
DO NOT SCALE PRINTS

PIEDMONT PACIFIC
GROOVE END FLEXIBLE PIPE COUPLING

TITLE:				
STYLE H COUPLING ASSEMBLY				
SCALE:	UNIT:	DRAWING NO.	SHEET:	REVISION:
N/A	INCH [MM]	1.0	1.0	9

Technical Proposal

14. MANDATORY/CRITICAL SPARE PARTS LIST FOR COMMISSIONING AND START-UP PHASES

(Ref.: P22-ET-09-000-001-0, Section 6 -Item “Suministro de Repuestos Mandatorios para Puesta en Marcha”)

Preliminary list of Mandatory/Critical Spares is presented in table below:

Manufacturer	Part #	Description	Qty.	Unit	Notes
Hardware					
Fil-Trek or Equivalent	tba	Cartridge Filter	1	set	
tba	tba	Fuses used in the panels	1	set	
Rosemount or Equivalent	tba	Pressure Transmitter	1	unit	Used in the HP system
Rosemount or Equivalent	tba	Conductivity sensor/electrode	1	set	Used on the line of final permeate
Flexitallic or Equivalent	tba	Seal/gasket kit	1	set	For HP Victaulic pipe
ABB or Equivalent	tba	Flowmeter	1	unit	
Ashcroft or Equivalent	tba	Pressure Gauges, SS304/316	3	unit	
IFM or Equivalent	tba	Level Transmitter	1	unit	

Technical Proposal

15. RECOMMENDED TWO-YEAR SPARE PARTS LIST

(Ref.: P22-ET-09-000-001-0, Section 6 -Item “Listado y Cotización Opcional de Repuestos para Dos Años”)

A Preliminary list of recommended 2-year Spares is presented in the table below:

Manufacturer	Part No.	Description	Qty.	Unit	Notes
Hardware					
Tampa Rubber EPDM Gaskets or Equivalent	Custom Part	Set of Gaskets (Low Pressure – Plastic Pipe)	4	set	
Flexitallic or Equivalent	Custom Part	Set pf Gaskets (High Pressure)	2	set	
Protec Arisawa	2081006	RO Vessel Head Assembly	1	unit	
Protec Arisawa	4080474-1	RO Vessel Head Locking Ring	3	unit	
Protec Arisawa	6100442MK	RO Vessel Seal Set	2	set	
Protec Arisawa	5080074	RO Membrane Adapters	4	unit	
Fedco or Equivalent	MSD-130 repair kit	SWRO High Pressure Pump Service Kit	1	set	
Pulsafeeder or Equivalent	LB02SA-XXXX (or equal)	Chemical Dosing Pump Repair Kit	2	set	
Foxboro or Equivalent	pH 10 (or equal)	pH Sensor	1	unit	
Foxboro or Equivalent	ORP 10 (or equal)	ORP Sensor	1	unit	
		Conductivity Sensor	1	unit	
Bray or Equivalent	Series 63, 120VAC	Solenoid Valve	2	unit	

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Manufacturer	Part No.	Description	Qty.	Unit	Notes
Ashcroft or Equivalent	Series 45, Monel	Pressure Gauge, Monel	2	unit	
Ashcroft or Equivalent	Series 45, SS	Pressure Gauge, SS304/316	2	unit	
Valves					
Bray or Equivalent	S01-0200	3" Lined Butterfly Valve	1	unit	
Asahi/America or Equivalent	Type 57L	2.5" Lined Butterfly Valve	1	unit	
Asahi/America or Equivalent	Type 57L	2" Lined Ball Valve	1	unit	
Asahi/America or Equivalent	Type 57P	4" Plastic Butterfly Valve	1	unit	
Asahi/America or Equivalent	Type 57P	3" Plastic Butterfly Valve	1	unit	
Asahi/America or Equivalent	Type 57P	2" Plastic Ball Valve	1	unit	
Hayward or Equivalent	TB Series	1.5 Plastic Ball Valve	5	unit	
Hayward or Equivalent	TB Series	1" Plastic Ball Valve	5	unit	

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16. CHEMICALS CONSUMPTION

Please refer to preliminary Chemicals Consumption below

Chemical Consumption Calculation									
1. Salinity Level (TDS):		43,000 - 53,000 mg/L							
2. Water Temperature:		19 - 24 °C							
3.UHPRO Feed Flowrate:		49 m ³ /hr = 1,176 m ³ /day = 215.6 gpm = 310464 gpd							
4. Total Product Flowrate:		21 m ³ /hr = 504 m ³ /day = 92.4 gpm =133056 gpd							
Item #	Chemical	For Use	Operation Mode	Dose	kg/year	kg/day	S.G(kg/L)	Frequency	Notes
1	Anti-Scalant (100%)	UHPRO	for Scale Inhibitor	0.5	215	0.6	1.031	24hr/day	
2	Citric Acid Cleaning(30%)	UHPRO	for UHPRO Acid CIP	20000ppm	152	0.4	1.01	1 time /3month	
3	NaOH Caustic Cleaning(50%)	UHPRO	for UHPRO Akali CIP	20000ppm	91	0.3	1.52	1 time /3month	

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17. POWER & LOAD CONSUMPTION LISTS

In alignment with Sections 3 and 19 of this proposal, the two attached power consumption documents serve distinct purposes:

SEC Guaranteed Value

This calculation includes only the power consumption of the RO High-Pressure (HP) Pump. All other equipment is excluded. This value will be used for the energy performance guarantee.

SEC Extended Value

This broader calculation includes the RO HP Pump along with additional equipment such as:

- Chemical dosing pumps
- CIP/Flushing pumps
- CIP heaters
- RO PLC power
- Other applicable auxiliary items

This extended value is not subject to energy performance guarantee requirements.

Both calculations are based on a Total Dissolved Solids (TDS) level of 53,000 ppm.

Note: At this bid stage, air conditioning (A/C) and internal/external lighting are not included in the SEC extended value calculation. The complete power consumption assessment, incorporating all applicable items and equipment, will be finalized during the project execution stage.

Please see attached calculations file

PRELIMINARY LOAD LISTS & POWER CONSUMPTIONS - SEC Guaranteed Value and SEC Extended Value

				PROJECT TITLE:	TALTAL Desal.Plant-BRINE RECOVERY SYSTEM									Production21.0m3/hr								PREPARED BY	CHECKED	APPROVED					PROJECT NO.		DOC NO.:	6501-E-CL-0001
				CLIENT:	AGUAS ANTOFAGASTA																	JR	ID	DN					6501		Rev. No.:	P1
				DOC. TITLE:	LOAD LIST & POWER CONSUMPTION - SEC Guaranteed Value																										DATE:	22-Jul-25
																								Electrical Loads								
Equipment ID No:	No. of Duty Pumps	No. of Standby Pump	No. in Intermittent Operation	Equipment Description	Flow	Inlet Pressure	Discharge Pressure	Fluid SG	Pump Effic.	Motor rating	Motor Service Factor	Motor Volts	Frequenc y	Number of Phases	Water	Motor Brake	Motor Loading	Motor Status		Motor Operating efficiency	VFD Efficiency	Consumed Kilowatts	Operating Power Factor	kVA	kVA Ampere	Operation Time Hr/Day	Consumed kWhr	Connected kW	StarterType			
					m3/h	Bar(g)	Bar(g)		%	in kW		Volts	Hz		Kilowatts	Kilowatts	%	Rating	S.F	%	%											
BRINE RECOVERY SYSTEM - Ultra High Pressure Reverse Osmosis Package																																
1	1	0	0	RO HP Pump	49.0	2	48.8	1.0	80.8	83.0	1.15	380	50	3	65	80.94	97.5	OK	OK	95.8	95.0	88.93	0.78	113.88	173.02	24	88.93	83.00	VFD			

Notes:

1. Other electrical loads will be provided in final document submittal.
2. Final kW ratings of pump-motors shall be determined from vendor data sheets.
3. Power consumption values are indicative of maximum 'not to exceed' values.

Total Consumed Power	93	kWhr
Total Connected Power	87	kW
Power Consumption	4.45	kW/m3
SEC Guaranteed Value		

				PROJECT TITLE:	AGUAS ANTOFAGASTA BRINE RECOVERY SYSTEM									Production21.0m3/hr								PREPARED BY	CHECKED	APPROVED					PROJECT NO.		DOC NO.:	6501-E-CL-0001
				CLIENT:	AGUAS ANTOFAGASTA																	JR	ID	DN					6501		Rev. No.:	P1
				DOC. TITLE:	LOAD LIST & POWER CONSUMPTION - SEC Extended Value																										DATE:	28-Jul-25
Equipment ID No:	No. of Duty Pumps	No. of Standby Pump	No. in Intermittent Operation	Equipment Description	Flow	Inlet Pressure	Discharge Pressure	Fluid SG	Pump Effic.	Motor rating	Motor Service Factor	Motor Volts	Frequenc y	Number of Phases	Water	Motor Brake	Motor Loading	Motor Status		Motor Operating efficiency	VFD Efficiency	Consumed Kilowatts	Operating Power Factor	kVA	kVA Ampere	Operation Time Hr/Day	Electrical Loads		StarterType			
					m3/h	Bar(g)	Bar(g)		%	in kW		Volts	Hz		Kilowatts	Kilowatts	%	Rating	S.F	%	%									Consumed kWhr	Connected kW	
BRINE RECOVERY SYSTEM - Ultra High Pressure Reverse Osmosis Package																																
1	1	0	0	RO HP Pump	49.0	2	48.8	1.0	80.8	83.0	1.15	380	50	3	65	80.94	97.5	OK	OK	95.8	95.0	88.93	0.78	113.88	173.02	24	88.93	83.00	VFD			
2	1	1	0	Antiscalant Dosing Pumps (UHPRO)						0.24	1.15	120	50	1			75.0	OK	OK	77.0		0.20	0.82	0.24	1.17	24	0.20	0.48	DOL			
3	1	0	0	CIP/Flushing Pump	54.5	0.25	4.38	1.0	82	9.3	1.15	380	50	3	6	7.63	82.0	OK	OK	95.8	95.0	88.93	0.85	104.74	159.14	0.02	0.07	9.3	VFD			
4	1	0	0	CIP Heater						16		380	50	3								16.00				0.02	15.20	16.0				
5	1	0	0	RO PLC (UPS powered by Others)						4.0		120	50	1								4.00				24	4.00	4.00	UPS Power			

Notes:

1. Other electrical loads will be provided in final document submittal.
2. Final kW ratings of pump-motors shall be determined from vendor data sheets.
3. Power consumption values are indicative of maximum 'not to exceed' values.
4. AC and the lighting realted power are excluded in current calculation. It will be provided during execution stage.

Total Consumed Power	114	kWhr
Total Connected Power	118	kW
Power Consumption	5.42	+/- 5% kW/m3
SEC Extended Value		

DOCUMENT No. 2 – COMMISSIONING & START- UP PLAN

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18. COMMISSIONING AND START-UP ACTIVITIES EXECUTION PLAN

(Ref.: Bases Administrativas Especiales (BAE), Section 20.1 – Document 2 “Plan de ejecución de actividades de comisionamiento y puesta en marcha”)

18.1 KEY PERSONNEL

All activities related to commissioning, start-up, on-site performance testing, and client personnel training will be led and directed by our **Field Services Manager, Mr. Andrew L. Fuller.**

Mr. Fuller brings over **25 years of experience** and a proven track record as an effective Project and Field Service Manager. He has successfully delivered complex water and wastewater treatment projects under technically demanding and time-constrained conditions, ensuring systems perform in accordance with their design specifications.

His extensive expertise in managing and executing on-site installation and commissioning of desalination and water/wastewater treatment systems will be a valuable asset throughout all phases of this project. Mr. Fuller will play a key role in guiding the project team toward selecting and implementing a technically sound and client-aligned solution.

As requested, Mr. Fuller's resume (curriculum vitae) is attached.



YEARS OF EXPERIENCE

25+

EDUCATION

- 1995/B. Eng.- Electrical Engineering / Kingston, Jamaica

MEMBERSHIPS & AFFILIATIONS

- S.M.E - Society for Mining, Metallurgy and Exploration

OTHER TRAINING

- Diploma / Supervisory Management

Mr. Fuller has over 25 years of experience in engineering design and project management, including start-up and commissioning of water projects. He possesses a solid theoretical and practical background, enabling him to handle various complex projects for municipal and industrial customers.

Mr. Fuller successfully utilizes his professional knowledge and rich experience to drive projects to successful completion, consistently meeting and/or exceeding client expectations.

DETAILED EXPERIENCE

BW Water Americas Inc.^(*), Tampa, FL – USA

Since he joined the company, held a few positions within the operation function

Project and Field Service Manager (Oct.2023 – Present)

Director of Project Management Operations (2012 – 2015)

Field Service Manager (Sept.2005 – 2012)

Project Manager (2000-Sept.2005)

- Manages and supervises field service technicians/engineers, providing training, support, and guidance, while ensuring employees and clients' operators are aware of potential safety issues related to the operation of equipment/systems.
- Oversees and ensures timely and high-quality service delivery, addressing customer issues and resolving complaints.
- Builds and maintains strong relationships with clients, ensuring their satisfaction and loyalty.
- During project execution, ensures project performance meets high standards, contract requirements, and stays within budget and on schedule. Manages the design, production, installation, and commissioning of membrane-based systems for public water supply and industrial process water requirements.
- Manages project budgets to maintain/improve profitability and efficiency.
- Possesses vast experience/expertise in executing and commissioning a wide variety of projects, that include seawater and brackish reverse osmosis systems, ultra-filtration systems, multimedia filters, granulated activated carbon treatment, and remineralization systems. Project sizes range from 0.25 to 50 MGD and they're located all over the world.

Major Projects

❖ *Municipal Client*

- Brunswick County's NWTP Expansion and Upgrades (Phase 3) - RO System 41 MGD - to be completed 2025
- Water Treatment Plant (WTP) Improvements - North Springs

- Improvement District (NSID) - Florida – 3 BWRO x 2.25MGD. 1 BWRO x 2MGD - Completed 2016| 2023
- Desalco's Point Lisas Desalination Plant - Expansion Projects - Trinidad & Tobago - 2 SWRO (1st pass) trains total capacity of 59,052 m3/day and 3 BWRO (2nd pass) trains total capacity of 68,136 m3/day. - Completed 2014 | 2016 | 2022
- Fargo Membrane WTP and Improvements North Dakota – RO systems 14 MGD - Completed 2019
- Doolittle Road WTP, Edinburg, Texas: BWRO System 2.0 MGD. Completed 2007.
- North Cameron County RO Plant, Texas: BWRO System 2.0 MGD. Completed 2007.
- La Sarah, Texas: BWRO System 1.0 MGD. Completed 2006
- Owassa WTP, Texas: BWRO System 3.0 MGD. Completed 2008.
- Southmost Regional Desalination Plant, Texas: BWRO System 6.0 MGD. Completed 2007.
- La Junta, Colorado: BWRO System 4.5 MGD. Completed December 2004
- Cooper City, Florida BWRO expansion, 4 MGD. Completed June 2003
- City of West Carrollton, Ohio: BWRO system, 2.0 MGD. Completed March 2005
- Town of Moundsville, West Virginia: BWRO, System, 3.0 MGD. Completed October 2007
- Village of Minster, Ohio: 2 MGD. Completed October 2007
- Long Beach Prototype, CA: NF-RO/SWRO, 0.3. Completed March 2005

❖ *Industrial Clients*

- PREPA_San Juan Power Plant -AWTP- Service Contract – Puerto Rico, 2.62MGD UF and 1.73MGD BWRO Systems - to be completed 2025
- Absolics-Chip Facility Project Georgia – WWTS: Sludge System - Completed 2024
- Pulmuone Food Facility- California - WWTP : DAF - Completed 2022
- TECO Bayside Power Station – Florida - Demineralizer System Upgrade (0.576 MGD Max) Completed 2023
- TECO- Polk Power Station-Reclaimed Water Project (TECO RTP) Original and Expansion 1 - Florida - BWRO Systems (3 x 2.4MGD) Completed 2015 | 2017
- Ammonia and Urea Plant, Mary, Turkmenistan: Clarifiers, R.O., Deionised systems, Mixed Bed Polishers for process and power plant. Completed in Sept 2014
- Grand Bahama Power Company, Grand Bahama: Containerized SWRO System. Completed March 2008

- Saqqara, Egypt: DMF, SWRO, UV, Demin, Deaerator for Oil Platform. Completed June 2008
- Altamira II, Tampico, Mexico: DMF, RO, Deaerator, Demin for Power plant boiler feed. Completed in November 2002

1995 - 2000: J. Wray and Nephew - Kingston, Jamaica.

- Overall responsibility for all plant maintenance (Electrical and Mechanical).
- Supervised the water team for process water production.
- Implemented annual budgets for plant maintenance, purchasing tools and equipment, and staff training.
- Introduced innovative ideas to improve plant equipment efficiency in line with company goals.
- Ensured the safety of operating equipment and maintained employee awareness of safety issues.
- Responsible for maintaining the ISO 9002 quality system for the Engineering section.

Note (*): Company name changes:

Nov.2023 - Present BW Water Americas Inc. | Jul 2017- Nov.2023 SafBon Water Technology Inc. | Sept 2015 – July 2017 Doosan Hydro Technology LLC/ Inc. | Before Sept.2015 American Engineering Service Inc.

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18.2 PRELIMINARY ON SITE FIELD SERVICE PROGRAM

To ensure the successful completion of commissioning and start-up activities, and to facilitate the subsequent achievement of **Provisional and Final Acceptance**, a final program and timeline will be developed and submitted following mutual agreement after system delivery.

Below is a **preliminary milestone schedule** based on current project requirements and conditions:

- **Delivery Period:** Maximum of **300 calendar days** after the **start date**
- **Installation Supervision, Commissioning and Start-Up, Testing, and Training Period:** Maximum of **21 calendar days**, initiated upon ADASA's formal request and confirmation by Adasa on Pre-check list that will state the required conditions and items needed by supplier (bidder) to go to site and initiate the on-site supervision services described in this proposal. This formal request from ADASA will occur within **180 calendar days** following receipt of the UHPRO system by ADASA.
- **On-Site Field Service Advance Notice:** ADASA will notify the supplier **30 calendar days in advance** of the required site appearance date.
- **Documentation and Close-Out:** within **30 calendar days** after completion of commissioning and start-up activities to submit all required documentation and fulfill all conditions necessary for **Provisional Acceptance (substantial completion)** of the order (BAE- Section 41)
- **Final Acceptance:** after warranty period

Please refer to the **Preliminary Project Execution Schedule** attached (see Section 11 of this proposal) for further details.

In accordance with Technical Specifications P22-ET-09-000-001, Sections 9 and 10, and the applicable Sections 9 to 13 of the Technical Report P22-IT-09-000-001, the On-Site Field Service Program will define the key steps, requirements for:

- Installation Supervision
- Commissioning & Start-up, including On-site performance testing to verify proper operation of the supplied system
- Training of ADASA personnel responsible for system operation
- Project close-out activities required to obtain Provisional Acceptance
- Warranty Period and Final Acceptance

All commissioning and start-up activities will be conducted at the Taltal Desalination Plant, located within ADASA's regional property at Guillermo Matta 221, Ciudad de Taltal, Región de Antofagasta.

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18.3 RESOURCES AND ASSIGNED SPECIALISTS TO BE USED

If our company is selected as the successful bidder for this project, Mr. Fuller will be responsible for selecting/assigning the field service team and coordinating with agreed and approved third parties (subvendors), if applicable, and the client to carry out all installation supervision, commissioning, start-up, on-site performance testing, and client training activities.

He will develop the final On-Site Field Service Program that was mentioned in section 18.2 of Proposal above where resources and assigned field service personnel and specialist will be identified:

Here is preliminary Resources and Assigned Specialists Plan for the On-Site Field Service Program/ supporting the installation supervision, commissioning, start-up performance testing, and operator training of a containerized Ultra High-Pressure Osmosis (UHPRO) system for brine recycle treatment ADASA project

❖ Resources and Assigned Specialists Plan (Preliminary-Sample Plan)

I. Project Scope Overview

This plan outlines the personnel and resources required to support the successful deployment and operational readiness of the containerized UHPRO system. The scope includes:

- Check list and verification of requirement to initiate any on-site field service program
- Installation supervision
- Mechanical and electrical Start-up
- Commissioning and performance testing
- Operator training and handover

II. Assigned Specialists and Roles

Role/Activity	Specialist	Responsibilities	Estimated Duration	Expected No. of people & No. trips
Field Service Manager	Principal in charge /Sr manager for field Svc. Scope (Andrew Fuller)	Overall coordination, client liaison, reporting. Compliance checks, risk assessments		1 person, 1Trip, if needed
Installation Supervision	Mechanical engineer and Electrical I&C engineer	Supervision of mechanical assembly, piping, and skid placement PLC integration, I/O checks, instrumentation setup	Within 9 Business Days	1 to 2 People and 1 trip

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Role/Activity	Specialist	Responsibilities	Estimated Duration	Expected No. of people & No. trips
Commissioning/ Start-up Performance testing and Operator Trainer	Field service Engineer	System flushing, pressure testing, membrane loading, performance validation, On-site training sessions, SOP handover, troubleshooting guidance	Within 12 business day	1 person and 1 trip

III. Support Resources

- Tools & Equipment xxx
- Calibration kits
- Flow meters and pressure gauges
- Electrical testing tools

IV. Documentation

- Pre-check of Onsite field services activities Accreditation fo personal as needed
- Schedule.
- Installation drawings
- Commissioning checklists
- SOPs and training manuals
- Performance test protocols
- Final & close up documentation for acceptance

V. Coordination with Client and Local Teams

- Weekly progress meetings with client representatives x
- Integration with local contractors for utilities and civil works

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DOCUMENT No. 3 – GUARANTEED SEC VALUE

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19. SIGNED DECLARATION OF THE SPECIFIC ENERGY CONSUMPTION (SEC) GUARANTEED VALUE

(Ref.: Bases Administrativas Especiales (BAE), Section 20.1 – Document 3 “Declaración de Consumo Específico de Energía Eléctrica Garantizado (CEE Garantizado))”

BW Water Americas Inc. (BWWA) has prepared Document 3 – **“Declaración de Consumo Específico de Energía Eléctrica Garantizado (CEE Garantizado)”** (in English Declaration of Guaranteed Specific Electrical Energy Consumption (CEE Guaranteed))

The signed Guaranteed SEC declaration letter is attached.



July 29 2025.

Aguas Antofagasta S.A. (ADASA)
CHILE

Subject: Ref. Licitación: 12803 SUMINISTRO MÓDULO RO SEGUNDA ETAPA PARA SALMUERA - PLANTA DESALADORA TALTAL (ex 12783) - Requirement BAE – 20.1 “Documentos Oferta Técnica” - Document 3: “Declaración De Consumo Específico De Energía Eléctrica Garantizado” (Declaration of Specific energy consumption)

Dear ADASA representative,

This letter serves to declare the Specific Energy Consumption (SEC) — in Spanish, Consumo específico de energía (CEp) — of our proposed UHPRO system, expressed in kWh/m³. This value corresponds to the guarantee offered for the operating condition with TDS in the range of 48,000–53,000 mg/L, as stated in Technical Specification ET-P22-ET-09-000-001, Clause 10.1.3.

Please note that the **guaranteed SEC value stated herein is based solely on the RO High-Pressure Pump of the RO Unit/Module**. The calculation does not include additional equipment such as chemical dosing pumps, CIP/Flushing pumps, CIP heaters, RO PLC power, or other auxiliary items as they outside of container which contains the cartridge filter, RO train, HP feed pump, energy recovery devices, A/C and Lighting.

Accordingly, for the purpose of this bid process stage, **the guaranteed SEC value is 4.45 kWh/m³**, based on the specified operating condition with TDS in the range of 48,000–53,000 mg/L. We understand that this value will be used to evaluate the commercial offer, provided we pass the technical evaluation stage.

Should you have any questions or require further clarification, please do not hesitate to contact us.

Sincerely,

 Digitally signed by
Andrew J Zaske
email= andrew.zaske@bw-water.
com
Date: 2025.07.29 19:09:40 EDT

Andrew J. Zaske
Vice President - Americas
Mobile No.: +1 612-409-2545
Andrew.Zaske@bw-water.com